

# FAI Governance Gaps and Future Derivation Work

Derivation Note D2.78 — Phase D2: Operational Variants and Decompositions

Sequential Number: #573

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## Attribution

This work derives from and formalizes the inter-Self coordination architecture introduced in *Inter-Self Coordination via Shared Substrate / Full Aspect Integration* (Li, April 2026), the third paper in the CKS theory series following *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026) and *The Instinct/Reasoning Separation Outside the Model* (Li, April 2026).

## Abstract

Phase D2 of the CKS derivation note series comprises eighty notes (#496–#575) formalizing operational prior art for Paper 3's architectural commitments across all six claims. D2.78 is a governance-honest capstone note, positioned two notes before Phase D2's close. It names seven coverage gaps in Phase D2's operational prior art: anti-pattern formalization depth (addressed by Phase D3), composition pair interactions (Phase D4), operational test suite expansion (Phase D5), boundary case depth (Phase D6), cross-paper inheritance edge formalization (Series CC), trilogy concept disambiguation (Series T), and Claim 6 empirical validation pathway (future academic work beyond the current derivation series). The note argues that naming these gaps, with specific planned resolutions, strengthens the prior-art position rather than weakening it: a corpus that is self-aware about its coverage scope is more credible than one that overclaims comprehensive treatment. The gaps identified here are the natural entry points for the phases that follow Phase D2. D2.79 and D2.80 complete Phase D2.

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## 1. Position Within the Derivation Series

D2.78 is the seventy-eighth note in Phase D2 and the five hundred and seventy-third note in the CKS derivation series overall. Phase D2 produces eighty operational prior-art notes covering Paper 3's six architectural claims: the shared substrate (Claim 1), Full Aspect Integration as the canonical operation over that substrate (Claim 2), the three-tier conflict-handling mechanism (Claim 3), the four-locus evolution-feed mechanism (Claim 4), configuration as substrate content with recursive applicability (Claim 5), and population-scale collective evolution as what the preceding five claims compose to at population register (Claim 6).

D2.78 is a governance-honest note. Its function is not to derive a further operational variant — that work is the province of D2.01 through D2.77 — but to survey what Phase D2 has built,

identify where its coverage is partial or deferred, and map each deferred area to the named phase that addresses it. D2.79 and D2.80 will complete Phase D2 with two final operational variant notes. D2.78 sits between the substantive derivation work and Phase D2's close.

The framing decision that governs this note is that gap acknowledgment and prior-art credibility are not in tension. A derivation corpus that names its coverage scope precisely — what it addresses, what it defers, and where the deferred work lives in the larger plan — is more defensible as prior art than a corpus that asserts comprehensive treatment without qualification. The gaps identified here are not admissions of incompleteness; they are the seams along which the subsequent phases are joined.

## **2. What Phase D2 Built**

Before naming the gaps, it is worth stating what Phase D2 accomplished. The eighty notes spanning #496–#575 formalized operational prior art for Paper 3 across a wide range of commitments and mechanisms.

For Claim 1, Phase D2 covered the shared substrate's perimeter-spanning construction, its governance-configured-persistence policy, its temporary-construction-and-dissolution architecture, and the property that all six Paper 1 commitments hold within its scope. For Claim 2, Phase D2 covered FAI's aspect-as-unit-of-exchange commitment, its n-ary cardinality floor at two-Self as the simplest instance, the full-merge-as-architectural-default commitment, the three pattern variants under orchestration-configured selection, and the exchange-bounded-to-reasoning-layer commitment that extends Paper 2's instinct/reasoning separation at the inter-Self boundary. For Claim 3, Phase D2 covered the three tiers of conflict handling — preserve as substrate-level state default, resolve via orchestration with rules as inspectable substrate content, escalate to humans across joint authority of participating Selves' governance — with particular attention to the cross-perimeter escalation shape as the genuinely fresh tier shape at Paper 3's scope. For Claim 4, Phase D2 covered the four-locus evolution-feed mechanism, including the hand-off at FAI dissolution, the per-mechanism feed structure, and the three-case layer-routing rule. For Claim 5, Phase D2 covered the configuration-as-substrate-content commitment and its recursive applicability, with the recursion bottoming at human-authored governance authority. For Claim 6, Phase D2 covered population-scale coordination as what the preceding five claims produce at population register, including the substrate-mediator ladder's six rungs and the joint-authority framing across population-level governance perimeters.

D2.24 produced a twenty-test operational test suite. D2.19 named twelve anti-patterns as governance failure modes. D2.37 established the minimum viable governance floor. D2.42 mapped scope boundaries. D2.47 covered asymmetric Selves. D2.68 touched on observational studies for Claim 6 validation.

## **3. Seven Coverage Gaps in Phase D2**

### **Gap 1 — Anti-Pattern Formalization Depth**

D2.19 named twelve FAI governance anti-patterns as failure modes across Paper 3's six claims. What D2.19 did not produce is the full prior-art depth that each anti-pattern warrants as a standalone topic: historical context locating the anti-pattern in adjacent literatures, detection criteria stating how a practitioner identifies the anti-pattern in a deployed system, and the argument structure that explains why the anti-pattern constitutes a governance failure rather than a legitimate architectural alternative. The naming was sufficient to establish the anti-patterns as prior art at the Phase D2 level; the depth is deferred.

**Phase D3 resolution (~30 notes, #576–#605):** Phase D3 will produce standalone anti-pattern formalization notes for each FAI governance failure mode, with full prior-art argumentation, historical context, and detection criteria, mirroring the anti-pattern formalization structure established in Phases A3 and B3 for Papers 1 and 2.

### **Gap 2 — Composition Pair Interactions**

Phase D2 covered how individual governance commitments operate. What Phase D2 did not extensively cover is how pairs of commitments interact when both are active simultaneously — the non-obvious dynamics that emerge when, for example, the minimum viable governance floor must coexist with high-frequency FAI event infrastructure, or when the three-tier conflict-handling mechanism operates over a substrate configured under Claim 5's recursive applicability commitment. Two commitments that each appear well-defined in isolation can create tension when both apply to the same event, the same substrate region, or the same governance decision.

**Phase D4 resolution (~30 notes, #606–#635):** Phase D4 will cover composition pairs — specific governance scenarios where two commitments must be applied simultaneously and their interaction is non-obvious. The scope includes composition pairs within Paper 3 and composition pairs that bridge Papers (Paper 1's conflict-as-first-class composed with Paper 3's three-tier mechanism; Paper 2's mating primitive composed with Paper 3's FAI as the inter-Self analog at broader scope).

### **Gap 3 — Systematic Operational Test Suite Expansion**

D2.24 produced a twenty-test operational test suite providing binary tests for Phase D2's most central governance commitments. Several commitments — particularly at the sub-commitment level within Claims 4 and 5, and at the population-scope level within Claim 6 — do not yet have corresponding binary tests. The test suite is a starting point, not a complete verification framework.

**Phase D5 resolution (~15 notes, #636–#650):** Phase D5 will expand the operational test suite to cover governance commitments that D2.24 did not address. Each Phase D5 note will produce standalone binary tests following the format established in D2.24 and in the corresponding Phase A5 and B5 test suite notes.

### **Gap 4 — Boundary Case Depth**

Phase D2 touched on boundary cases at several points: D2.42 mapped scope boundaries at the inter-Self perimeter, D2.37 addressed the minimum viable governance floor, and D2.47 covered asymmetric Selves. These notes established that boundary cases exist and that the architecture has principled responses to them. What they did not produce is dedicated boundary case notes for every edge condition at Paper 3's commitments — particularly the edge conditions at Claim 6's population-scale scope.

**Phase D6 resolution (~15 notes, #651–#665):** Phase D6 will produce standalone boundary case notes for FAI governance at the edges of its commitments, mirroring the boundary case structure established in Phases A6 and B6. Each note will take one edge condition and develop the architectural response in full, establishing that Paper 3's commitments extend coherently to the boundary.

### **Gap 5 — Cross-Paper Inheritance Edge Formalization**

Phase D2 derived Paper 3's operational prior art primarily from a Paper 3-internal perspective. When a Paper 3 commitment inherits from a Paper 1 or Paper 2 commitment, Phase D2 noted the inheritance but did not produce standalone notes formalizing each inheritance edge as its own prior-art contribution. The inheritance edges between Paper 3 and Papers 1 and 2 are numerous: the shared substrate's perimeter-spanning property inherits from Paper 1's substrate commitment; FAI's exchange-bounded-to-reasoning-layer commitment inherits from Paper 2's instinct/reasoning separation extended at the inter-Self boundary; the preserve tier inherits from Paper 1's conflict-as-first-class commitment; configuration-as-substrate-content inherits from Paper 1's human-governed commitment. Each of these inheritance edges merits standalone formalization.

**Series CC resolution (~40 notes, #666–#705):** Series CC will produce standalone cross-paper inheritance notes for Paper 3. Phase CC.1 (~20 notes, #666–#685) formalizes Paper 3 ↔ Paper 1 inheritance edges; Phase CC.2 (~20 notes, #686–#705) formalizes Paper 3 ↔ Paper 2 inheritance edges. Each CC note states the inheritance relationship explicitly, identifies what Paper 3 adds at inter-Self scope, and provides the prior-art argument without requiring re-defense of the parent commitment.

### **Gap 6 — Trilogy Disambiguation Series T**

The CKS trilogy uses a set of concepts that appear across Papers 1, 2, and 3 with scope-specific meanings. The trilogy ambiguity map catalogs fifteen such concepts — substrate, Self, aspect, conflict, orchestration, governance, evolution, perimeter, and others — each of which operates at a different scope in each paper while using the same term. Phase D2 used the trilogy ambiguity map as a reference when drafting notes but did not produce standalone disambiguation notes for any of the fifteen concepts. Standalone disambiguation notes establish, as prior art, the precise scope-specific meaning of each term at each paper level — protecting against misreadings that collapse Paper 3's inter-Self usage into Paper 1's cell-scope or Paper 2's intra-Self usage.

**Series T resolution (~50 notes, #706–#755):** Series T will produce standalone disambiguation notes for each trilogy concept across five phases: T1 (cross-paper concept disambiguation, ~15

notes), T2 (substrate-mediator ladder rung transitions, ~10 notes), T3 (trilogy-wide composition, ~10 notes), T4 (trilogy-wide anti-patterns, ~10 notes), and T5 (trilogy synthesis, ~5 notes).

### Gap 7 — Claim 6 Empirical Validation Pathway

D2.68 addressed observational studies as one approach to empirical evidence for Claim 6 — the extension claim that population-scale collective evolution is what Paper 3's five preceding claims produce at population register. The note established that a population of governed Selves producing FAI events over time is, in principle, an observable phenomenon. What D2.68 did not produce is a formalized research methodology for population-scale governance research: a research design with specified populations, measurement instruments, comparison conditions, and analysis procedures that could validate or challenge Claim 6's population-scope assertion in a publishable empirical study.

**Future academic work (beyond the current derivation series scope):** Formal empirical validation of Claim 6 is future academic work. The derivation series establishes the architectural prior art; an empirical program addressing whether accumulated FAI events across governed populations produce the collective evolution structures Claim 6 describes requires a separate research effort with appropriate institutional context. This is acknowledged here as a gap in the current series' scope, not as a deficiency in the prior-art argument.

## 4. Summary Table

| Gap | Description                      | Resolution                       | Phase(s)             | Notes                           |
|-----|----------------------------------|----------------------------------|----------------------|---------------------------------|
| 1   | Anti-pattern formalization depth | Standalone anti-pattern notes    | D3 (~30, #576–#605)  | Full depth per failure mode     |
| 2   | Composition pair interactions    | Dedicated composition pair notes | D4 (~30, #606–#635)  | Intra-P3 and cross-paper        |
| 3   | Operational test suite expansion | Additional binary test notes     | D5 (~15, #636–#650)  | Extends D2.24's 20 tests        |
| 4   | Boundary case depth              | Standalone boundary case notes   | D6 (~15, #651–#665)  | Edge conditions, all claims     |
| 5   | Cross-paper inheritance edges    | Standalone CC inheritance notes  | CC (~40, #666–#705)  | CC.1: P3↔P1; CC.2: P3↔P2        |
| 6   | Trilogy concept disambiguation   | Standalone T series notes        | T (~50, #706–#755)   | T1–T5 sub-phases                |
| 7   | Claim 6 empirical validation     | Future research program          | Beyond current scope | Research gap, not prior-art gap |

## 5. Governance-Honest Positioning as Prior-Art Strength

The conventional framing of coverage gaps treats them as liabilities — areas where a corpus falls short of what it sets out to do. This note inverts that framing. In the context of a defensive-publication derivation series, naming coverage gaps with precision, and providing a concrete plan for resolving each, is a demonstration of governance at the corpus level, not a concession.

A prior-art corpus that asserts comprehensive coverage without qualification invites the objection that its coverage claims are overclaims. A prior-art corpus that maps its own scope — here are the eighty notes in Phase D2, here is what they cover, here are seven areas where the coverage is lighter than subsequent phases will provide, here is the phase that addresses each —

provides a roadmap that a reader can follow and verify. The gaps enumerated in §3 are not silences in the prior-art record; they are annotated boundaries between Phase D2 and the phases that follow it.

This self-awareness is consistent with the governance posture the CKS series establishes in its source papers. Paper 1's human-governed commitment includes the right to inspect any content at any time; D2.78 applies that posture to the derivation corpus itself, making the corpus's coverage structure inspectable. Paper 3's configuration-as-substrate-content commitment, with recursive applicability, means that the rules governing the architecture are themselves subject to governance. D2.78 applies the analog: the rules governing what the derivation corpus covers are themselves stated explicitly, subject to inspection and refinement. The prior-art record is strengthened, not weakened, by this transparency.

## **6. Phase D2 Near Completion: What D2.79 and D2.80 Complete**

D2.78 is the seventy-eighth of eighty Phase D2 notes. Two notes remain: D2.79 and D2.80. These final two notes will complete Phase D2's operational prior art with two further operational variant notes in the standard derivation format, addressing areas identified in the derivation plan that D2.78's governance-honest positioning falls outside.

With D2.79 and D2.80 complete, Phase D2 closes and Phase D3 begins. The seven gaps identified in §3 are the natural entry points for the phases that follow. Phase D3 inherits the twelve anti-patterns D2.19 named and develops each into a standalone prior-art formalization. Phase D4 inherits the individual governance commitments Phase D2 established and formalizes the pairs. Phase D5 inherits D2.24's twenty-test suite and expands it. Phase D6 inherits the boundary-adjacent notes Phase D2 produced and extends them into dedicated boundary case treatments. Series CC inherits Phase D2's inheritance references and elevates each to standalone status. Series T inherits the trilogy ambiguity map and produces the disambiguation notes the map anticipated.

Phase D2 is, on this view, not merely a phase that established prior art for Paper 3's commitments. It is also the phase that built the foundation — the named anti-patterns, the initial test suite, the boundary-adjacent notes, the inheritance references — on which Phases D3 through D6 and Series CC and T will stand. The gaps are not absences; they are the joints.

## **7. Conclusion**

Phase D2 produced eighty operational prior-art notes for Paper 3's inter-Self coordination architecture, covering all six claims at a level of operational depth that establishes prior art for the core mechanisms. D2.78 names seven areas where Phase D2's coverage is lighter than the subsequent dedicated phases will provide: anti-pattern formalization depth (Phase D3), composition pair interactions (Phase D4), operational test suite expansion (Phase D5), boundary case depth (Phase D6), cross-paper inheritance edge formalization (Series CC), trilogy concept disambiguation (Series T), and the Claim 6 empirical validation pathway (future academic work beyond the current series).

Naming these gaps is a governance act, not a concession. The derivation corpus applies to itself the same standard of inspectability it establishes for the architectures it derives from. Two notes remain in Phase D2. The phases that follow are already planned.

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*End of D2.78.*